COM111 Tutorial 2a

1. For the circuit in Fig. 1, find Vo/Vs in terms of α, R1, R2, R3, and R4. If R1 = R2 = R3 = R4, what value of α will produce |Vo/Vs| = 10?

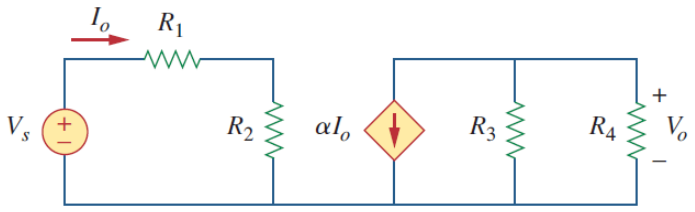


Fig. 1

2. Find *R*eq for the circuit in Fig. 2.

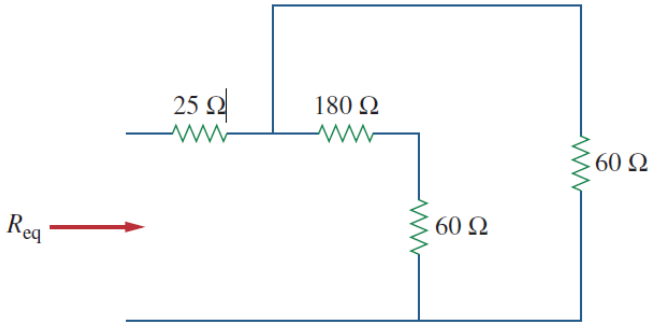


Fig. 2

3. For the circuit in Fig. 3, determine *i*1 to *i*5.

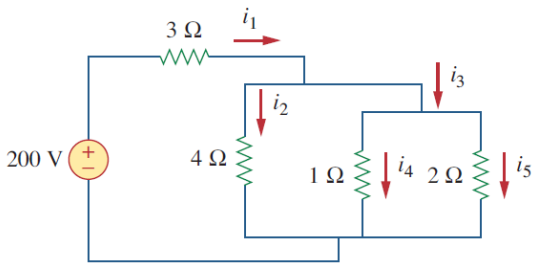


Fig. 3

4. Find *i*1, *i*2, *i*3, and *i*4 in the circuit in Fig. 4

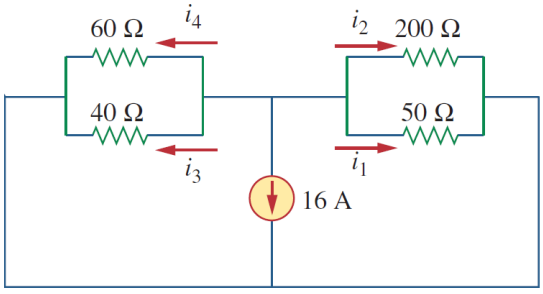


Fig. 4

5. Calculate *Vo* and *Io* in the circuit of Fig. 5.

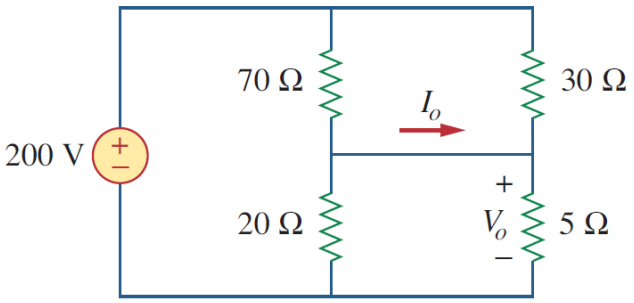


Fig. 5

6. Find *R*eq and *io* in the circuit of Fig. 6.

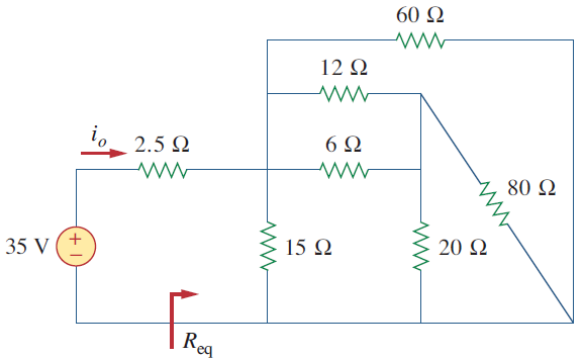


Fig. 6

7. If *R*eq = 50 Ω in the circuit of Fig. 7, find *R*.

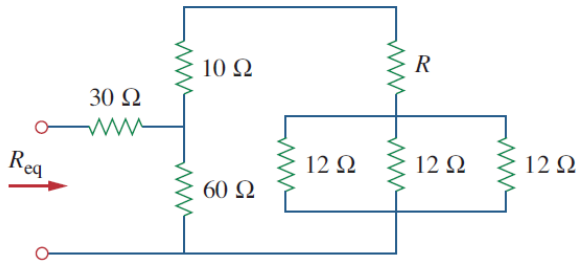


Fig. 7

8. Find the equivalent resistance *R*ab in the circuit of Fig. 8.

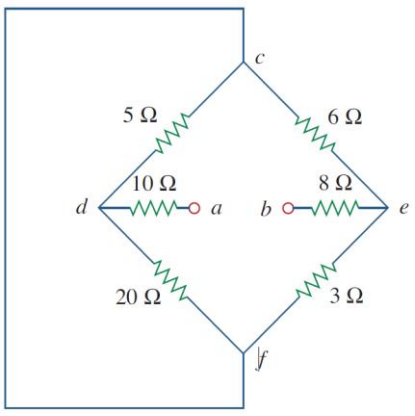


Fig. 8

9. Use the voltage-division principle to calculate the voltage(s) in Fig. 9 (a) and (b).

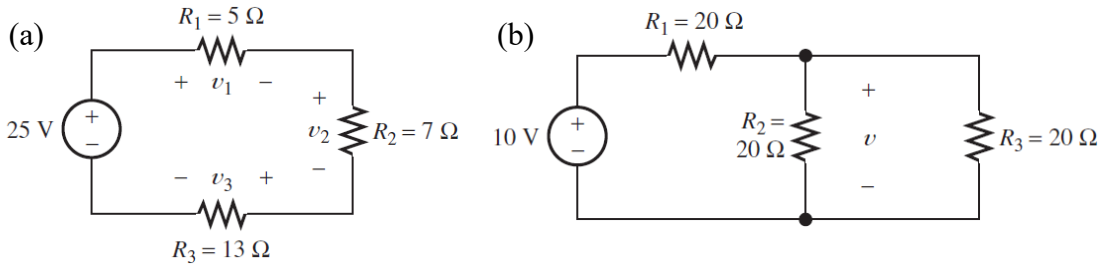


Fig. 9

10. Use the current-division principle to calculate current(s) in Fig. 10 (a) and (b).

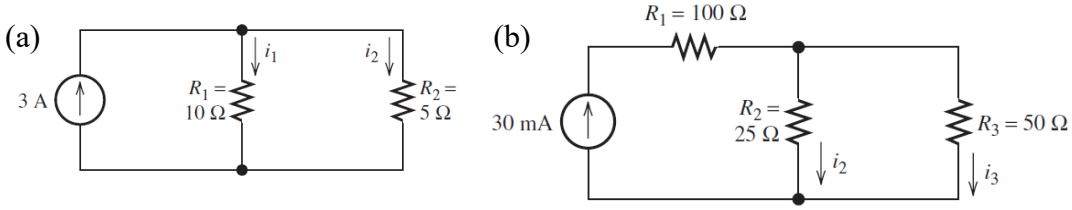


Fig. 10

11. A parallel circuit (i.e., all elements are in parallel with one another) has a 60 Ω resistance, a 20 Ω resistance, an unknown resistance Rx, and 30 mA current source. The current through the unknown resistance is 10 mA. Determine the value of Rx.